AEC - Q100 - Rev-G
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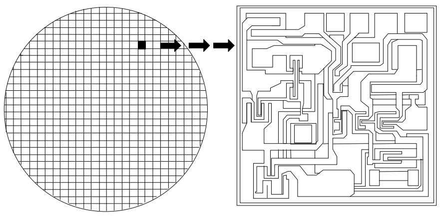
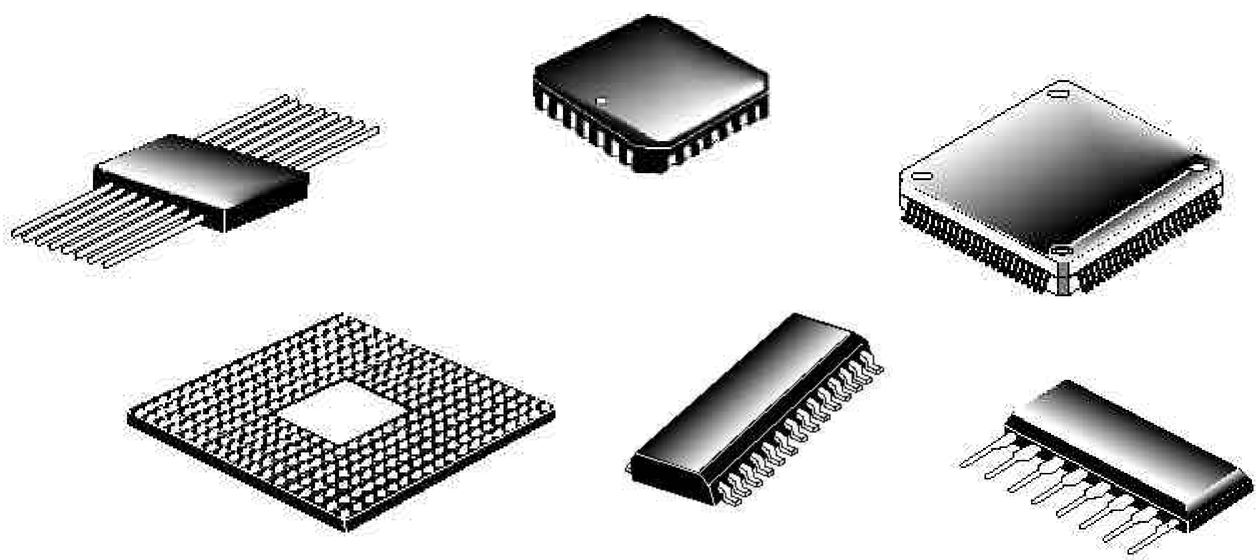


FAILURE MECHANISM BASED STRESS TEST QUALIFICATION FOR INTEGRATED CIRCUITS



Automotive Electronics Council
Component Technical Committee

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Attachments

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Acknowledgment

Any document involving a complex technology brings together experience and skills from many sources. The Automotive Electronics Council would especially like to recognize the following significant contributors to the revision of this document:

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Automotive Electronics Council
Component Technical Committee

FAILURE MECHANISM BASED STRESS TEST QUALIFICATION FOR PACKAGED INTEGRATED CIRCUITS

Text enhancements and differences made since the last revision of this document are shown as underlined areas. Several figures and tables have also been revised, but changes to these areas have not been underlined.

1. SCOPE

This document contains a set of failure mechanism based stress tests and defines the minimum stress test driven qualification requirements and references test conditions for qualification of integrated circuits (ICs). These tests are capable of stimulating and precipitating semiconductor device and package failures. The objective is to precipitate failures in an accelerated manner compared to use conditions. This set of tests should not be used indiscriminately. Each qualification project should be examined for:

a. Any potential new and unique failure mechanisms.
b. Any situation where these tests/conditions may induce failures that would not be seen in the application.
c. Any extreme use condition and/or application that could adversely reduce the acceleration.

Use of this document does not relieve the IC supplier of their responsibility to meet their own company's internal qualification program. In this document, "user" is defined as all customers using a device qualified per this specification. The user is responsible to confirm and validate all qualification data that substantiates conformance to this document. Supplier usage of the device temperature grades as stated in this specification in their part information is strongly encouraged.

1.1 Purpose

The purpose of this specification is to determine that a device is capable of passing the specified stress tests and thus can be expected to give a certain level of quality/reliability in the application.

1.2 Reference Documents

Current revision of the referenced documents will be in effect at the date of agreement to the qualification plan. Subsequent qualification plans will automatically use updated revisions of these referenced documents.

1.2.1 Automotive

AEC-Q001 Guidelines for Part Average Testing
AEC-Q002 Guidelines for Statistical Yield Analysis
AEC-Q003 Guidelines for Characterizing the Electrical Performance
AEC-Q004 Zero Defects Guideline (DRAFT)
SAE J1752/3 Integrated Circuits Radiated Emissions Measurement Procedure

1.2.2 Military

MIL-STD-883 Test Methods and Procedures for Microelectronics

1.2.3 Industrial

JEDEC JESD-22 Reliability Test Methods for Packaged Devices
EIA/JESD78 IC Latch-Up Test
UL-STD-94 Tests for Flammability of Plastic materials for parts in Devices and Appliances
IPC/JEDEC J-STD-020 Moisture/Reflow Sensitivity Classification for Plastic Integrated Circuit Surface Mount Devices
JESD89 Measurement and Reporting of Alpha Particle and Terrestrial Cosmic Ray-Induced Soft Errors in Semiconductor Devices
JESD89-1 System Soft Error Rate (SSER) Test Method
JESD89-2 Test Method For Alpha Source Accelerated Soft Error Rate
JESD89-3 Test Method for Beam Accelerated Soft Error Rate

1.3 Definitions

1.3.1 AEC Q100 Qualification

Successful completion and documentation of the test results from requirements outlined in this document allows the supplier to claim that the part is “AEC Q100 qualified”. The supplier, in agreement with the user, can perform qualification at sample sizes and conditions less stringent than what this document requires. However, that part cannot be considered “AEC Q100 qualified” until such time that the unfulfilled requirements can be completed.

1.3.2 Approval for Use in an Application

"Approval" is defined as user approval for use of a part in their application. The user's method of approval is beyond the scope of this document.

1.3.3 Definition of Part Operating Temperature Grade

The part operating temperature grades are defined below:

Grade 0: -40°C to +150°C ambient operating temperature range
Grade 1: -40°C to +125°C ambient operating temperature range
Grade 2: -40°C to +105°C ambient operating temperature range
Grade 3: -40°C to +85°C ambient operating temperature range
Grade 4: 0°C to +70°C ambient operating temperature range

2. GENERAL REQUIREMENTS

2.1 Objective

The objective of this specification is to establish a standard that defines operating temperature grades for integrated circuits based on a minimum set of qualification requirements.

2.1.1 Zero Defects

Qualification and some other aspects of this document are a subset of, and contribute to, the achievement of the goal of Zero Defects. Elements needed to implement a zero defects program can be found in AEC-Q004 Zero Defects Guideline.

2.2 Precedence of Requirements

In the event of conflict in the requirements of this standard and those of any other documents, the following order of precedence applies:

a. The purchase order
b. The individual device specification
c. This document
d. The reference documents in section 1.2 of this document
e. The supplier's data sheet

For the device to be considered a qualified part per this specification, the purchase order and/or the individual device specification cannot waive or detract from the requirements of this document.

2.3 Use of Generic Data to Satisfy Qualification and Requalification Requirements

2.3.1 Definition of Generic Data

The use of generic data to simplify the qualification process is strongly encouraged. Generic data can be submitted to the user as soon as it becomes available to determine the need for any additional testing. To be considered, the generic data must be based on a matrix of specific requirements associated with each characteristic of the device and manufacturing process as shown in Table 3 and Appendix 1**. If the generic data contains any failures, the data is not usable as generic data unless the supplier has documented and implemented corrective action or containment for the failure condition that is acceptable to the user.**

Appendix 1 defines the criteria by which components are grouped into a qualification family for the purpose of considering the data from all family members to be equal and generically acceptable for the qualification of the device in question. For each stress test, two or more qualification families can be combined if the reasoning is technically sound (i.e., supported by data).

Table 1: Part Qualification/Requalification Lot Requirements

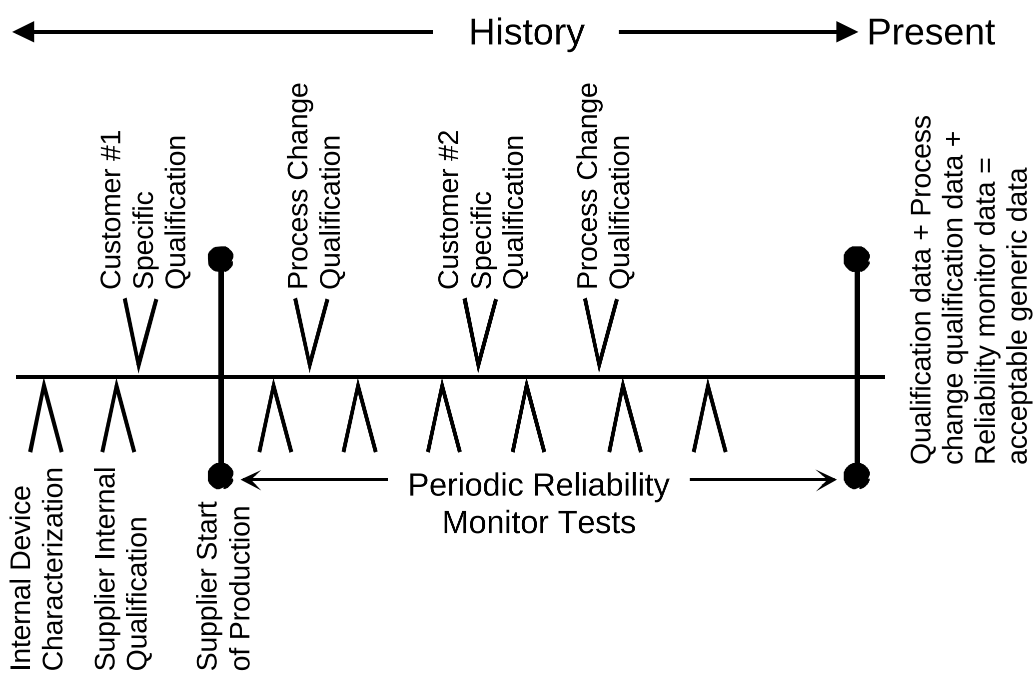
Part Information	Lot Requirements for Qualification
New device, no applicable generic data.	Lot and sample size requirements per Table 2.
A part in a family is qualified. The part to be qualified is less complex and meets the Family Qualification Definition per Appendix 1.	Only device specific tests as defined in section 4.2 are required. Lot and sample size requirements per Table 2 for the required tests.
A new part that has some applicable generic data.	Review Appendix 4 to determine required tests from Table 2. Lot and sample sizes per Table 2 for the required tests.
Part process change.	Review Table 3 to determine which tests from Table 2 are required. Lot and sample sizes per Table 2 for the required tests.
Part was environmentally tested to all the test extremes, but was electrically end-point tested at a temperature less than the Grade required.	The electrical end-point testing on at least 1 lot (that completed qualification testing) must meet or exceed the temperature extremes for the device Grade required. Sample sizes shall be per Table 2.
Qualification/Requalification involving multiple sites.	Refer to Appendix 1, section 3.
Qualification/Requalification involving multiple families.	Refer to Appendix 1, section 3.

With proper attention to these qualification family guidelines, information applicable to other devices in the family can be accumulated. This information can be used to demonstrate generic reliability of a device family and minimize the need for device-specific qualification test programs. This can be achieved through qualification and monitoring of the most complex (e.g., more memory, A/D, larger die size) device in the qualification family and applying this data to less complex devices that subsequently join this family. Sources of generic data should come from supplier-certified test labs, and can include internal supplier's qualifications, cell structure/standard circuit characterization and testing, user-specific qualifications, and supplier's in-process monitors. The generic data to be submitted must meet or exceed the test conditions specified in Table 2. **End-point test temperatures must address the worst case temperature extremes for the device operating temperature grade being qualified on at least one lot of data.** Failure to do so will result in the supplier testing 1 lot or, if there is no applicable or acceptable existing generic data, 3 lots for the stress test(s) in question on the device to be qualified. **The user(s) will be the final authority on the acceptance of generic data in lieu of test data.**

Table 3 defines a set of qualification tests that must be considered for any changes proposed for the component. The Table 3 matrix is the same for both new processes and requalification associated with a process change. This table is a superset of tests that the supplier and user should use as a baseline for discussion of tests that are required for the qualification in question. **It is the supplier's responsibility to present rationale for why any of the recommended tests need not be performed.**

2.3.3 Time Limit for Acceptance of Generic Data

There are no time limits for the acceptability of generic data as long as all reliability data taken since the initial qualification is submitted to the user for evaluation. This data must come from the specific part or a part in the same qualification family, as defined in Appendix 1. This data includes any customer specific data (if customer is non-AEC, withhold customer name), process change qualification, and periodic reliability monitor data (see Figure 1).



Note: Some process changes (e.g., die shrink) will affect the use of generic data such that data obtained before these types of changes will not be acceptable for use as generic data.

Figure 1: Generic Data Time Line

2.4 Test Samples

2.4.1 Lot Requirements

Test samples shall consist of a representative device from the qualification family. Where multiple lot testing is required due to a lack of generic data, test samples as indicated in Table 2 must be composed of approximately equal numbers from non-consecutive wafer lots, assembled in non-consecutive molding lots. That is, they must be separated in the fab or assembly process line by at least one non-qualification lot.

2.4.2 Production Requirements

All qualification devices shall be produced on tooling and processes at the manufacturing site that will be used to support part deliveries at production volumes. Other electrical test sites may be used for electrical measurements after their electrical quality is validated.

2.4.3 Reusability of Test Samples

Devices that have been used for nondestructive qualification tests may be used to populate other qualification tests. Devices that have been used for destructive qualification tests may not be used any further except for engineering analysis.

2.4.4 Sample Size Requirements

Sample sizes used for qualification testing and/or generic data submission must be consistent with the specified minimum sample sizes and acceptance criteria in Table 2.

If the supplier elects to use generic data for qualification, the specific test conditions and results must be recorded and available to the user (preferably in the format shown in Appendix 4). Existing applicable generic data should first be used to satisfy these requirements and those of section 2.3 for each test requirement in Table 2. Device specific qualification testing should be performed if the generic data does not satisfy these requirements.

2.4.5 Pre- and Post-stress Test Requirements

End-point test temperatures (room, hot and/or cold) are specified in the "Additional Requirements" column of Table 2 for each test. The specific value of temperature must address the worst case operating temperature grade extremes on at least one lot of data (generic or device specific) submitted per test. For example, if a supplier designs a device intended solely for use in an operating temperature Grade 3 environment (e.g., -40°C to +85°C), the end-point test temperature extremes need only address those application limits. Qualification to applications in higher operating temperature grade environments (e.g., -40°C to +125°C for Grade 1) will require testing of at least one lot using these additional end-point test temperature extremes.

2.5 Definition of Test Failure After Stressing

Test failures are defined as those devices not meeting the individual device specification, criteria specific to the test, or the supplier's data sheet, in the order of significance as defined in section 2.2. Any device that shows external physical damage attributable to the environmental test is also considered a failed device. If the cause of failure is agreed (by the manufacturer and the user) to be due to mishandling, ESD, or some other cause unrelated to the test conditions, the failure shall be discounted, but reported as part of the data submission.

3. QUALIFICATION AND REQUALIFICATION

3.1 Qualification of a New Device

The stress test requirement flow for qualification of a new device is shown in Figure 2 with the corresponding test conditions defined in Table 2. For each qualification, the supplier must have data available for all of these tests, whether it is stress test results on the device to be qualified or acceptable generic data. A review shall also be made of other devices in the same generic family to ensure that there are no common failure mechanisms in that family. Justification for the use of generic data, whenever it is used, must be demonstrated by the supplier and approved by the user.

For each device qualification, the supplier must have available the following:

- Certificate of Design, Construction and Qualification (see Appendix 2)
- Stress Test Qualification data (see Table 2 & Appendix 4)
- Data indicating the level of fault grading of the software used for qualification (when applicable to the device type) per Q100-007 that will be made available to the customer upon request

3.1.1 Qualification of a New Device Manufactured in a Currently Qualified Family

If justified by the supplier and agreed to by the user, new or redesigned products (die revisions) manufactured in a currently qualified family may be qualified using one (1) wafer/assembly lot.

3.2 Requalification of a Changed Device

Requalification of a device is required when the supplier makes a change to the product and/or process that impacts (or could potentially impact) the form, fit, function, quality and/or reliability of the device (see Table 3 for guidelines).

3.2.1 Process Change Notification

The supplier will meet the user requirements for product/process changes.

3.2.2 Changes Requiring Requalification

As a minimum, any change to the product, as defined in Appendix 1, requires performing the applicable tests listed in Table 2, using Table 3 to determine the requalification test plan. Table 3 should be used as a guide for determining which tests are applicable to the qualification of a particular part change or whether equivalent generic data can be submitted for that test(s).

3.2.3 Criteria for Passing Requalification

All requalification failures shall be analyzed for root cause, with corrective and preventive actions established as required. **The device and/or qualification family may be granted "qualification status" if, as a minimum, proper containment is demonstrated and approved by the user, until corrective and preventative actions are in place.**

3.2.4 User Approval

A change may not affect a device's operating temperature grade, but may affect its performance in an application. Individual user authorization of a process change will be required for that user's particular application(s), and this method of authorization is outside the scope of this document.

4. QUALIFICATION TESTS

4.1 General Tests

Test flows are shown in Figure 2 and test details are given in Table 2. Not all tests apply to all devices. For example, certain tests apply only to ceramic packaged devices, others apply only to devices with NVM, and so on. The applicable tests for the particular device type are indicated in the "Note" column of Table 2. The "Additional Requirements" column of Table 2 also serves to highlight test requirements that supersede those described in the referenced test method. Any unique qualification tests or conditions requested by the user and not specified in this document shall be negotiated between the supplier and user requesting the test.

4.2 Device Specific Tests

The following tests must be performed on the specific device to be qualified for all hermetic and plastic packaged devices. Generic data is not allowed for these tests. Device specific data, if it already exists, is acceptable.

1. Electrostatic Discharge (ESD) - All product.

2. Latch-up (LU) - All product.

3. Electrical Distribution - The supplier must demonstrate, over the operating temperature grade, voltage, and frequency ranges, that the device is capable of meeting the parametric limits of the device specification. This data must be taken from at least three lots, or one matrixed (or skewed) process lot, and must represent enough samples to be statistically valid, see Q100-009. It is strongly recommended that the final test limits be established using AEC-Q001 Guidelines For Part Average Testing.

4. Other Tests - A user may require other tests in lieu of generic data based on his experience with a particular supplier.

4.3 Wearout Reliability Tests

Testing for the failure mechanisms listed below must be available to the user whenever a new technology or material relevant to the appropriate wearout failure mechanism is to be qualified. The data, test method, calculations, and internal criteria need not be demonstrated or performed on the qualification of every new device, but should be available to the user upon request.

- Electromigration
- Time-Dependent Dielectric Breakdown (or Gate Oxide Integrity Test) - for all MOS technologies
- Hot Carrier Injection - for all MOS technologies below 1 micron
- Negative Bias Temperature Instability
- Stress Migration

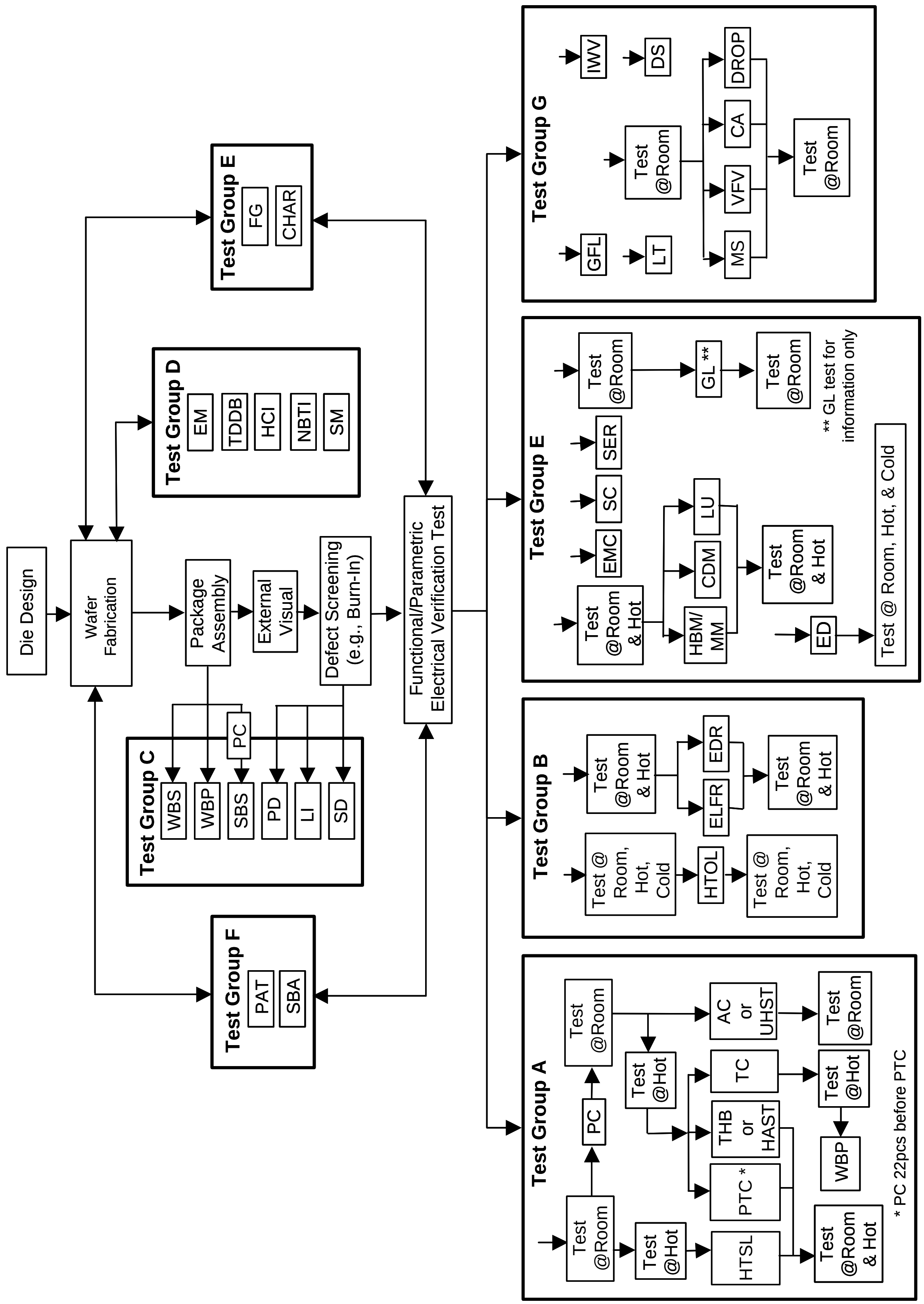


Figure 2: Qualification Test Flow

Table 2: Qualification Test Methods

TEST GROUP A – ACCELERATED ENVIRONMENT STRESS TESTS								
STRESS	**ABV**	**#**	**NOTES**	**SAMPLE SIZE / LOT**	**NUMBER OF LOTS**	**ACCEPT CRITERIA**	**TEST METHOD**	**ADDITIONAL REQUIREMENTS**
Preconditioning	**PC**	A1	P, B, S, N, G	77	3	0 Fails	JEDEC J-STD-020 JESD22-A113	Performed on surface mount devices only. PC performed before THB/HAST, AC/UHST, TC, and PTC stresses. It is recommended that J-STD-020 be performed to determine what preconditioning level to perform in the actual PC stress per JA113. The minimum acceptable level for qualification is level 3 per JA113. Where applicable, preconditioning level and Peak Reflow Temperature must be reported when preconditioning and/or MSL is performed. Delamination from the die surface in JA113/J-STD-020 is acceptable if the device passes the subsequent Qualification tests. Any replacement of devices must be reported. **TEST before and after PC at room temperature.**
Temperature-Humidity-Bias or Biased HAST	**THB or HAST**	A2	P, B, D, G	77	3	0 Fails	JEDEC JESD22-A101 or A110	For surface mount devices, PC before THB (85°C/85%RH for 1000 hours) or HAST (130°C/85%RH for 96 hours, or 110°C/85%RH for 264 hours). **TEST before and after THB or HAST at room and hot temperature.**
Autoclave or Unbiased HAST or Temperature-Humidity (without Bias)	**AC or UHST or TH**	A3	P, B, D, G	77	3	0 Fails	JEDEC JESD22-A102, A118, or A101	For surface mount devices, PC before AC (121°C/15psig for 96 hours) or unbiased HAST (130°C/85%RH for 96 hours, or 110°C/85%RH for 264 hours). For packages sensitive to high temperatures and pressure (e.g., BGA), PC followed by TH (85°C/85%RH) for 1000 hours may be substituted. **TEST before and after AC or UHST at room temperature.**

Table 2: Qualification Test Methods (continued)

TEST GROUP A – ACCELERATED ENVIRONMENT STRESS TESTS (CONTINUED)								
STRESS	**ABV**	**#**	**NOTES**	**SAMPLE SIZE / LOT**	**NUMBER OF LOTS**	**ACCEPT CRITERIA**	**TEST METHOD**	**ADDITIONAL REQUIREMENTS**
Temperature Cycling	**TC**	A4	H, P, B, D, G	77	3	0 Fails	JEDEC JESD22-A104 and Appendix 3	PC before TC for surface mount devices. **Grade 0:** -65ºC to +175ºC for 500 cycles, -50ºC to +175ºC for 1000 cycles, or -50ºC to +150ºC for 2000 cycles. **Grade 1:** -65°C to +150°C for 500 cycles or-50°C to +150°C for 1000 cycles. **Grade 2:** -50ºC to +150ºC for 500 cycles or-50ºC to +125ºC for 1000 cycles. **Grade 3:** -50ºC to +125ºC for 500 cycles or-50ºC to +105ºC for 1000 cycles. **Grade 4:** -10ºC to +105ºC for 500 cycles or-10ºC to + 90ºC for 1000 cycles. **TEST before and after TC at hot temperature.** After completion of TC, decap five devices from one lot and perform WBP (test #C2) on corner bonds (2 bonds per corner) and one mid-bond per side ***on each device***. Preferred decap procedure to minimize damage and chance of false data is shown in Appendix 3.
Power Temperature Cycle	**PTC**	A5	H, P, B, D, G	45	1	0 FAILS	JEDEC JESD22-A105	PC 22pcs before PTC for surface mount devices. Test required only on devices with maximum rated power = 1 watt or ΔT_J = 40ºC or devices designed to drive inductive loads. **Grade 0:** T_a of -40ºC to +150ºC for 1000 cycles. **Grade 1:** T_a of -40ºC to +125ºC for 1000 cycles. **Grades 2 to 4:** T_a -40ºC to +105ºC for 1000 cycles. Thermal shut-down shall not occur during this test. **TEST before and after PTC at room and hot temperature.**
High Temperature Storage Life	**HTSL**	A6	H, P, B, D, G, K	45	1	0 FAILS	JEDEC JESD22-A103	**Plastic Packaged Parts** **Grade 0:** +175ºC for 1000 hours or +150ºC for 2000 hours. **Grade 1:** +150ºC for 1000 hours or +175ºC for 500 hours. **Grades 2 to 4:** +125ºC for 1000 hours or +150ºC for 500 hours. **Ceramic Packaged Parts** +250ºC for 10 hours or +200ºC for 72 hours. **TEST before and after HTSL at room and hot temperature.** *** NOTE: Data from Test B3 (EDR) can be substituted for Test A6 (HTSL) if package and grade level requirements are met.**

Table 2: Qualification Test Methods (continued)

TEST GROUP B – ACCELERATED LIFETIME SIMULATION TESTS								
STRESS	**ABV**	**#**	**NOTES**	**SAMPLE SIZE / LOT**	**NUMBER OF LOTS**	**ACCEPT CRITERIA**	**TEST METHOD**	**ADDITIONAL REQUIREMENTS**
High Temperature Operating Life	**HTOL**	B1	H, P, B, D, G, K	<u>77</u>	<u>3</u>	0 FAILS	JEDEC JESD22-A108	For devices containing NVM, endurance preconditioning must be performed before HTOL per Q100-005. **Grade 0:** +175ºC T_a for 408 hours or +150ºC T_a for 1000 hours. **Grade 1:** +150°C T_a for 408 hours or +125°C T_a for 1000 hours. **Grade 2:** +125ºC T_a for 408 hours or +105ºC T_a for 1000 hours. **Grade 3:** +105ºC T_a for 408 hours or +85ºC T_a for 1000 hours. **Grade 4:** +90ºC T_a for 408 hours or +70ºC T_a for 1000 hours. V_{cc} (max) at which dc and ac parametrics are guaranteed. Thermal shut-down shall not occur during this test. **TEST before and after HTOL at room, hot, and cold temperature.**
Early Life Failure Rate	**ELFR**	B2	H, P, B, N, G	800	3	0 FAILS	AEC Q100-008	Devices that pass this stress can be used to populate other stress tests. Generic data is applicable. **TEST before and after ELFR at room and hot temperature.**
NVM Endurance, Data Retention, and Operational Life	**EDR**	B3	H, P, B, D, G, <u>K</u>	77	3	0 FAILS	AEC Q100-005	**TEST before and after EDR at room and hot temperature.**

Table 2: Qualification Test Methods (continued)

STRESS	ABV	#	NOTES	SAMPLE SIZE / LOT	NUMBER OF LOTS	ACCEPT CRITERIA	TEST METHOD	ADDITIONAL REQUIREMENTS
TEST GROUP C – PACKAGE ASSEMBLY INTEGRITY TESTS								
Wire Bond Shear	**WBS**	C1	H, P, D, G	30 bonds from a minimum of 5 devices		C_{PK} >1.33 P_{Pk} >1.67	AEC Q100-001	At appropriate time interval for each bonder to be used.
Wire Bond Pull	**WBP**	C2	H, P, D, G			C_{PK} >1.33 P_{Pk} >1.67 or 0 Fails after TC (test #A4)	MIL-STD883 Method 2011	Condition C or D. <u>For Au wire diameter ≥1mil, m</u>inimum pull strength after TC = 3 grams. <u>For Au wire diameter <1mil, refer to Figure 2011-1 in MIL-STD-883 Method 2011 as a guideline for minimum pull strength. For Au wire diameter <1mil, wire bond pull shall be performed with the hook over the ball bond and not at mid-wire.</u>
Solderability	**SD**	C3	H, P, D, G	15	1	>95% lead coverage	JEDEC JESD22-B102	If burn-in screening is normally performed on the device before shipment, samples for SD must first undergo burn-in. Perform 8 hour steam aging prior to testing (1 hour for Au-plated leads).
Physical Dimensions	**PD**	C4	H, P, B, D, G	10	3	C_{PK} >1.33 P_{PK} >1.67	JEDEC JESD22-B100 and B108	See applicable JEDEC standard outline and individual device spec for significant dimensions and tolerances.
Solder Ball Shear	**SBS**	C5	B	5 balls from a min. of 10 devices	3	C_{PK} >1.33 P_{PK} >1.67	AEC Q100-010	PC thermally (two 220ºC reflow cycles) before integrity (mechanical) testing.
Lead Integrity	**LI**	C6	H, P, D, G	10 leads from each of 5 parts	1	No lead breakage or cracks	JEDEC JESD22-B105	Not required for surface mount devices. Only required for through-hole devices.
TEST GROUP D – DIE FABRICATION RELIABILITY TESTS								
STRESS	**ABV**	**#**	**NOTES**	**SAMPLE SIZE / LOT**	**NUMBER OF LOTS**	**ACCEPT CRITERIA**	**TEST METHOD**	**ADDITIONAL REQUIREMENTS**
Electromigration	**EM**	D1	---	---	---	---	---	The data, test method, calculations and internal criteria should be available to the user upon request for new technologies.
Time Dependent Dielectric Breakdown	**TDDB**	D2	---	---	---	---	---	The data, test method, calculations and internal criteria should be available to the user upon request for new technologies.

Table 2: Qualification Test Methods (continued)

TEST GROUP D – DIE FABRICATION RELIABILITY TESTS (CONTINUED)								
STRESS	**ABV**	**#**	**NOTES**	**SAMPLE SIZE / LOT**	**NUMBER OF LOTS**	**ACCEPT CRITERIA**	**TEST METHOD**	**ADDITIONAL REQUIREMENTS**
Hot Carrier Injection	**HCI**	D3	---	---	---	---	---	The data, test method, calculations and internal criteria should be available to the user upon request for new technologies.
Negative Bias Temperature Instability	**NBTI**	D4	---	---	---	---	---	The data, test method, calculations and internal criteria should be available to the user upon request for new technologies.
Stress Migration	**SM**	D5	---	---	---	---	---	The data, test method, calculations and internal criteria should be available to the user upon request for new technologies.
TEST GROUP E – ELECTRICAL VERIFICATION TESTS								
STRESS	**ABV**	**#**	**NOTES**	**SAMPLE SIZE / LOT**	**NUMBER OF LOTS**	**ACCEPT CRITERIA**	**TEST METHOD**	**ADDITIONAL REQUIREMENTS**
Pre- and Post-Stress Function/Parameter	**TEST**	E1	H, P, B, N, G	All	All	0 Fails	Test program to supplier data sheet or user specification	Test is performed as specified in the applicable stress reference and the additional requirements in Table 2 and illustrated in Figure 2. Test software used shall meet the requirements of Q100-007. All electrical testing before and after the qualification stresses are performed to the limits of the individual device specification in temperature and limit value.
Electrostatic Discharge Human Body Model / Machine Model	**HBM / MM**	E2	H, P, B, D	See Test Method	1	0 Fails 2KV HBM (H2 or better) 200V MM (M3 or better)	AEC Q100-002 Q100-003	**TEST before and after ESD at room and hot temperature**. At least one of these models must be performed. Device shall be classified according to the maximum withstand voltage level. Device levels <2000V HBM and/or <200V MM require specific user approval.
Electrostatic Discharge Charged Device Model	**CDM**	E3	H, P, B, D	See Test Method	1	0 Fails 750V corner pins, 500V all other pins (C3B or better)	AEC Q100-011	**TEST before and after ESD at room and hot temperature**. Device shall be classified according to the maximum withstand voltage level. Device levels <750V corner pins and/or <500V all other pins CDM require specific user approval.

Table 2: Qualification Test Methods (continued)

TEST GROUP E – ELECTRICAL VERIFICATION TESTS (CONTINUED)								
STRESS	**ABV**	**#**	**NOTES**	**SAMPLE SIZE / LOT**	**NUMBER OF LOTS**	**ACCEPT CRITERIA**	**TEST METHOD**	**ADDITIONAL REQUIREMENTS**
Latch-Up	**LU**	E4	H, P, B, D	6	1	0 Fails	AEC Q100-004	See attached procedure for details on how to perform the test. **TEST before and after LU at room and hot temperature.**
Electrical Distributions	**ED**	E5	H, P, B, D	30	3	See AEC Q100-009	AEC Q100-009	Supplier and user to mutually agree upon electrical parameters to be measured and accept criteria. **TEST at room, hot, and cold temperature.**
Fault Grading	**FG**	E6	---	---	---	See Section 4 of AEC Q100-007	AEC Q100-007	For production testing, see section 4 of Q100-007 for test requirements.
Characterization	**CHAR**	E7	---	---	---	---	AEC Q003	To be performed on new technologies and part families.
Electrothermally-Induced Gate Leakage	**GL**	E8	D, P, B, S	6	1	0 Fails	AEC Q100-006	For information only. **TEST before and after GL at room temperature within 96 hours of GL stress completion.**
Electromagnetic Compatibility	**EMC**	E9	---	1	1	---	SAE J1752/3 – Radiated Emissions	See Appendix 5 for guidelines on determining the applicability of this test to the device to be qualified. This test and its accept criteria is performed per agreement between user and supplier on a case-by-case basis.
Short Circuit Characterization	**SC**	E10	D, G	10	3	0 Fails	AEC Q100-012	**Applicable to all smart power devices.** This test and statistical evaluation (see section 4 of Q100-012) shall be performed per agreement between user and supplier on a case-by-case basis.
Soft Error Rate	**SER**	E11	H, P, D, G	3	1	---	JEDEC Un-accelerated: JESD89-1 or Accelerated: JESD89-2 & JESD89-3	**Applicable to devices with memory sizes ^{3}1Mbit SRAM or DRAM based cells.** Either test option (un-accelerated or accelerated) can be performed, in accordance to the referenced specifications. This test and its accept criteria is performed per agreement between user and supplier on a case-by-case basis. Final test report shall include detailed test facility location and altitude data.

Table 2: Qualification Test Methods (continued)

TEST GROUP F – DEFECT SCREENING TESTS

STRESS	ABV	#	NOTES	SAMPLE SIZE / LOT	NUMBER OF LOTS	ACCEPT CRITERIA	TEST METHOD	ADDITIONAL REQUIREMENTS
Process Average Testing	**PAT**	F1	---	---	---	---	AEC Q001	It is highly desirable that the supplier adopt these tests in their standard manufacturing operation.
Statistical Bin/Yield Analysis	**SBA**	F2	---	---	---	---	AEC Q002	

TEST GROUP G – CAVITY PACKAGE INTEGRITY TESTS

STRESS	ABV	#	NOTES	SAMPLE SIZE / LOT	NUMBER OF LOTS	ACCEPT CRITERIA	TEST METHOD	ADDITIONAL REQUIREMENTS
Mechanical Shock	**MS**	G1	H, D, G	39	3	0 Fails	JEDEC JESD22-B104	Y1 plane only, 5 pulses, 0.5 msec duration, 1500 g peak acceleration. **TEST before and after at room temperature.**
Variable Frequency Vibration	**VFV**	G2	H, D, G	39	3	0 Fails	JEDEC JESD22-B103	20 Hz to 2 KHz to 20 Hz (logarithmic variation) in >4 minutes, 4X in each orientation, 50 g peak acceleration. **TEST before and after at room temperature.**
Constant Acceleration	**CA**	G3	H, D, G	39	3	0 Fails	MIL-STD-883 Method 2001	Y1 plane only, 30 K g-force for <40 pin packages, 20 K g-force for 40 pins and greater. **TEST before and after at room temperature.**
Gross/Fine Leak	**GFL**	G4	H, D, G	39	3	0 Fails	MIL-STD-883 Method 1014	Any single-specified fine test followed by any single-specified gross test. For ceramic packaged cavity devices only.
Package Drop	**DROP**	G5	H, D, G	5	1	0 Fails	---	Drop part on each of 6 axes once from a height of 1.2m onto a concrete surface. This test is for MEMS cavity devices only. **TEST before and after DROP at room temperature.**
Lid Torque	**LT**	G6	H, D, G	5	1	0 Fails	MIL-STD-883 Method 2024	For ceramic packaged cavity devices only.
Die Shear	**DS**	G7	H, D, G	5	1	0 Fails	MIL-STD-883 Method 2019	To be performed before cap/seal for all cavity devices.
Internal Water Vapor	**IWV**	G8	H, D, G	3	1	0 Fails	MIL-STD-883 Method 1018	For ceramic packaged cavity devices only.

Legend for Table 2

Notes:
- **H** Required for hermetic packaged devices only.
- **P** Required for plastic packaged devices only.
- **B** Required Solder Ball Surface Mount Packaged (BGA) devices only.
- **N** Nondestructive test, devices can be used to populate other tests or they can be used for production.
- **D** Destructive test, devices are not to be reused for qualification or production.
- **S** Required for surface mount plastic packaged devices only.
- **G** Generic data allowed. See section 2.3, Table 1, and Appendix 1.
- **K** Use method AEC-Q100-005 for preconditioning a stand-alone Non-Volatile Memory integrated circuit or an integrated circuit with a Non-Volatile Memory module.

Reference Number for the particular test.

* All electrical testing before and after the qualification stresses are performed to the limits of the individual device specification in temperature and limit value.

Table 3: Process Change Qualification Guidelines for the Selection of Tests

A2 Temperature Humidity Bias or HAST
A3 Autoclave or Unbiased HAST
A4 Temperature Cycling
A5 Power Temperature Cycling
A6 High Temperature Storage Life
B1 High Temperature Operating Life
B2 Early Life Failure Rate
B3 NVM Endurance, Data Retention
C1 Wire Bond Shear
C2 Wire Bond Pull
C3 Solderability
C4 Physical Dimensions
C5 Solder Ball Shear
C6 Lead Integrity
D1 Electromigration
D2 Time Dependent Dielectric Breakdown
D3 Hot Carrier Injection
<u>D4 Negative Bias Temperature Instability</u>
<u>D5 Stress Migration</u>
E2 Human Body / Machine Model ESD
E3 Charged Device Model ESD
E4 Latch-up
E5 Electrical Distribution
E7 Characterization
E8 Gate Leakage
E9 Electromagnetic Compatibility
<u>E10 Short Circuit Characterization</u>
<u>E11 Soft Error Rate</u>
G1-4 Mechanical <u>Series</u>
G5 Package Drop
G6 Lid Torque
G7 Die Shear
G8 Internal Water Vapor

Note: A letter or "●" indicates that performance of that stress test should be **considered** for the appropriate process change

Table 2 Test #	**A2**	**A3**	**A4**	**A5**	**A6**	**B1**	**B2**	**B3**	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**D1**	**D2**	**D3**	**<u>D4</u>**	**<u>D5</u>**	**E2**	**E3**	**E4**	**E5**	**E7**	**E8**	**E9**	**<u>E10</u>**	**<u>E11</u>**	**G1-G4**	**G5**	**G6**	**G7**	**G8**
Test Abbreviation	**THB**	**AC**	**TC**	**PTC**	**HTSL**	**HTOL**	**ELFR**	**EDR**	**WBS**	**WBP**	**SD**	**PD**	**SBS**	**LI**	**EM**	**TDDB**	**HCI**	**<u>NBTI</u>**	**<u>SM</u>**	**HBM / MM**	**CDM**	**LU**	**ED**	**CHAR**	**GL**	**EMC**	**<u>SC</u>**	**<u>SER</u>**	**<u>MECH</u>**	**DROP**	**LT**	**DS**	**IWV**
DESIGN																																	
Active Element Design		●	●	M		●	●	DJ							D	D	D	D	D	●	●	●	●	●	S	●	●	●		F			
Circuit Rerouting			A	M																●	●	●	●	●	S	●	●						
Wafer Dimension / Thickness			E	M		●	●		E	E								●		E	E	E	●										
WAFER FAB																																	
Lithography	●		●	M		●	G		●	●								●					●										
Die Shrink	●	●		M		●	●	DJ							●	●	●	●	●	●	●	●	●	●	S	●	●	●					
Diffusion/Doping				M		●	G											●		●	●	●	●	●	S								
Polysilicon			●	M		●		DJ										●		●	●	●	●	●	S								
Metallization / Vias / Contacts	●	●	●	M		●			●	●					●				●				●	●	S		●						
Passivation / Oxide / Interlevel Dielectric	K	K	●	M		●	GN	DJ	K	●						●	●	●	●	●	●	●	●	●	PS								
Backside Operation			●	M		●														M	M	●		●					H			H	
FAB Site Transfer	●	●	●	M		●	●	J	●	●					●	●	●	●	●	●	●	●	●		S				H			H	
ASSEMBLY																																	
Die Overcoat / Underfill	●	●	●	M	●	●																			S			●					H
Leadframe Plating	●	●	●	M	●					C	●			●																		H	
Bump Material / Metal System	●	●	●	M	●	●						●	●															●					
Leadframe Material		●	●	M	●					●	●	●		●													●		H			H	
Leadframe Dimension		●	●	M							●	●		●													●		H				
Wire Bonding		●	●	Q	●				●	●													M				●		H				
Die Scribe/Separate	●	●	●	M																													
Die Preparation / Clean	●	●		M		●			●	●																						H	
Package Marking											B																						
Die Attach	●	●	●	M		●																	●				●		H			H	H
Molding Compound	●	●	●	M	●	●	●				●	●		●											S			●					
Molding Process	●	●	●	M	●	●					●	●		●											S								
Hermetic Sealing		H	H		H							H		H															H		H		H
New Package	●	●	●	M	●	●	●		●	●	●	●	T	●						●	●		●		S		●		H			H	H
Substrate / Interposer	●	●	●	M	●	●			●	●			T												S				H			H	H
Assembly Site Transfer	●	●	●	M		●	●		●	●	●	●	T	●									●		S				H			H	H

A Only for peripheral routing
B For symbol rework, new cure time, temp
C If bond to leadfinger
D Design rule change
E Thickness only
F MEMS element only
G Only from non-100% burned-in parts
H Hermetic only
J EPROM or E^2PROM
K Passivation only
M For devices requiring PTC
N Passivation and gate oxide
P Passivation and interlevel dielectric
Q Wire diameter decrease
S <u>For</u> plastic SMD only
T <u>For</u> Solder Ball SMD <u>only</u>

Appendix 1: Definition of a Qualification Family

The qualification of a particular process will be defined within, but not limited to, the categories listed below. The supplier will provide a complete description of each process and material of significance. There must be valid and obvious links between the data and the subject of qualification.

For devices to be categorized in a qualification family, they all must share the same major process and materials elements as defined below. All devices using the same process and materials are to be categorized in the same qualification family for that process and are qualified by association when one family member successfully completes qualification with the exception of the device specific requirements of section 4.2. Prior qualification data obtained from a device in a specific family may be extended to the qualification of subsequent devices in that family.

For broad changes that involve multiple attributes (e.g., site, materials, processes), refer to section A1.3 of this appendix and section 2.3 of Q100, which allows for the selection of worst-case test vehicles to cover all the possible permutations.

A1.1 Fab Process

Each process technology (e.g., CMOS, NMOS, Bipolar, etc.) must be considered and qualified separately. No matter how similar, processes from one fundamental fab technology cannot be used for another. For BiCMOS devices, data must be taken from the appropriate technology based on the circuit under consideration.

Family requalification with the appropriate tests is required when the process or a material is changed (see Table A1 for guidelines). The important attributes defining a qualification family are listed below:

a. **Wafer Fab Technology** (e.g., CMOS, NMOS, Bipolar, etc.)

b. **Wafer Fab Process** - consisting of the same attributes listed below:
   - Circuit element feature size (e.g., layout design rules, die shrinks, contacts, gates, isolations)
   - Substrate (e.g., orientation, doping, epi, wafer size)
   - Number of masks (supplier must show justification for waiving this requirement)
   - Lithographic process (e.g., contact vs. projection, E-beam vs. X-ray, photoresist polarity)
   - Doping process (e.g., diffusion vs. ion implantation)
   - Gate structure, material and process (e.g., polysilicon, metal, salicide, wet vs. dry etch)
   - Polysilicon material, thickness range and number of levels
   - Oxidation process and thickness range (for gate and field oxides)
   - Interlayer dielectric material and thickness range
   - Metallization material, thickness range and number of levels
   - Passivation material and thickness range
   - Die backside preparation process and metallization

c. **Wafer Fab Site**

A1.2 Assembly Process - Plastic or Ceramic

The processes for plastic and ceramic package technologies must be considered and qualified separately. For devices to be categorized in a qualification family, they all must share the same major process and material elements as defined below. Family requalification with the appropriate tests is required when the process or a material is changed. The supplier must submit technical justification to the user to support the acceptance of generic data with pin (ball) counts, die sizes, substrate dimensions/material/thickness, paddle sizes and die aspect ratios different than the device to be qualified. The important attributes defining a qualification family are listed below:

a. **Package Type** (e.g., DIP, SOIC, PLCC, QFP, PGA, PBGA)
   - Same cross-sectional dimensions (Width x Height)
   - Range of paddle (flag) size (maximum and minimum dimensions) qualified for the die size/aspect ratio under consideration
   - Substrate base material (e.g., PBGA)

b. **Assembly Process** - consisting of the same attributes listed below:
   - Leadframe base material
   - Leadframe plating (internal and external to the package)
   - Die attach material
   - Wire bond material, wire diameter, presence of downbonds, and process
   - Plastic mold compound, organic substrate material, or ceramic package material
   - Solder Ball metallization system (if applicable)
   - Heatsink type, material, dimensions

c. **Assembly Site**

A1.3 Qualification of Multiple Sites and Families

A1.3.1 Multiple Sites

When the specific product or process attribute to be qualified or requalified will affect more than one wafer fab site or assembly site, a minimum of one lot of testing per affected site is required.

A1.3.2 Multiple Families

When the specific product or process attribute to be qualified or requalified will affect more than one wafer fab family or assembly family, the qualification test vehicles should be: 1) One lot of a single device type from each of the families that are projected to be most sensitive to the changed attribute, or 2) Three lots total (from any combination of acceptable generic data and stress test data) from the most sensitive families if only one or two families exist.

Below is the recommended process for qualifying changes across many process and product families:

a. Identify all products affected by the proposed process changes.

b. Identify the critical structures and interfaces potentially affected by the proposed process change.

c. Identify and list the potential failure mechanisms and associated failure modes for the critical structures and interfaces (see the example in Table A1). Note that steps (a) to (c) are equivalent to the creation of an FMEA.

d. Define the product groupings or families based upon similar characteristics as they relate to the structures and device sensitivities to be evaluated, and provide technical justification for these groupings.

e. Provide the qualification test plan, including a description of the change, the matrix of tests and the representative products, that will address each of the potential failure mechanisms and associated failure modes.

f. Robust process capability must be demonstrated at each site (e.g., control of each process step, capability of each piece of equipment involved in the process, equivalence of the process step-by-step across all affected sites) for each of the affected process steps.

Table A1: Example of Failure Mode/Mechanism List for a Passivation Change

Critical Structure or Interface	Potential Failure Mechanism	Associated Failure Modes	On These Products
Passivation to Mold Compound Interface	Passivation Cracking - Corrosion	Functional Failures	All Die
	Mold Compound - Passivation Delamination	Corner Wire Bond Failures	Large Die
Passivation to Metallization Interface	Stress-Induced Voiding	Functional Failures	Die with Minimum Width Metal Lines
	Ionic Contamination	Leakage, Parametric Shifts	All Die
Polysilicon and Active Resistors	Piezoelectric Leakage	Parametric Shifts (e.g., Resistance, Gain, Offset)	Analog Products

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Appendix 2: Q100 Certification of Design, Construction and Qualification

Supplier Name: **Date:**

The following information is required to identify a device that has met the requirements of AEC-Q100. Submission of the required data in the format shown below is optional. **All entries must be completed; if a particular item does not apply, enter "Not Applicable".** This template can be downloaded from the AEC website at http://www.aecouncil.com.

This template is available as a stand-alone document.

Item Name	Supplier Response
1. User's Part Number:	
2. Supplier's Part Number/Data Sheet:	
3. Device Description:	
4. Wafer/Die Fab Location & Process ID: a. Facility name/plant #: b. Street address: c. Country:	 _____ _____ _____
5. Wafer Probe Location: a. Facility name/plant #: b. Street address: c. Country:	 _____ _____ _____
6. Assembly Location & Process ID: a. Facility name/plant #: b. Street address: c. Country:	 _____ _____ _____
7. Final Quality Control A (Test) Location: a. Facility name/plant #: b. Street address: c. Country:	 _____ _____ _____
8. Wafer/Die: a. Wafer size: b. Die family: c. Die mask set revision & name: d. Die photo:	 _____ See attached ☐ Not available ☐
9. Wafer/Die Technology Description: a. Wafer/Die process technology: b. Die channel length: c. Die gate length: d. Die supplier process ID (Mask #): e. Number of transistors or gates: f. Number of mask steps:	 _____
10. Die Dimensions: a. Die width: b. Die length: c. Die thickness (finished):	
11. Die Metallization: a. Die metallization material(s): b. Number of layers: c. Thickness (per layer): d. % of alloys (if present):	
12. Die Passivation: a. Number of passivation layers: b. Die passivation material(s): c. Thickness(es) & tolerances:	

13. Die Overcoat Material (e.g., Polyimide):	_____
14. Die Cross-Section Photo/Drawing:	See attached ☐ Not available ☐
15. Die Prep Backside: a. Die prep method: b. Die metallization: c. Thickness(es) & tolerances:	
16. Die Separation Method: a. Kerf width (µm): b. Kerf depth (if not 100% saw): c. Saw method:	 _____ _____ Single ☐ Dual ☐
17. Die Attach: a. Die attach material ID: b. Die attach method: c. Die placement diagram:	 See attached ☐ Not available ☐
18. Package: a. Type of package (e.g., plastic, ceramic, unpackaged): b. Ball/lead count: c. JEDEC designation (e.g., MS029, MS034, etc.): d. Lead (Pb) free (< 0.1% homogenous material): e. Package outline drawing:	 _____ Yes ☐ No ☐ See attached ☐ Not available ☐
19. Mold Compound: a. Mold compound supplier & ID: b. Mold compound type: c. Flammability rating: d. Fire Retardant type/composition: e. Tg (glass transition temperature)(°C): f. CTE (above & below Tg)(ppm/°C):	 UL 94 V1 ☐ UL 94 V0 ☐ _____ CTE1 (above Tg) = _____ CTE2 (below Tg) = _____
20. Wire Bond: a. Wire bond material: b. Wire bond diameter (mils): c. Type of wire bond at die: d. Type of wire bond at leadframe: e. Wire bonding diagram:	 _____ _____ See attached ☐ Not available ☐
21. Leadframe (if applicable): a. Paddle/flag material: b. Paddle/flag width (mils): c. Paddle/flag length (mils): d. Paddle/flag plating composition: e. Paddle/flag plating thickness (µinch): f. Leadframe material: g. Leadframe bonding plating composition: h. Leadframe bonding plating thickness (µinch): i. External lead plating composition: j. External lead plating thickness (µinch):	

22. Substrate (if applicable): a. Substrate material (e.g., FR5, BT, etc.): b. Substrate thickness (mm): c. Number of substrate metal layers: d. Plating composition of ball solderable surface: e. Panel singulation method: f. Solder ball composition: g. Solder ball diameter (mils):	 _____ _____ _____ _____ _____ _____ _____
23. Unpackaged Die (if not packaged): a. Under Bump Metallurgy (UBM) composition: b. Thickness of UBM metal: c. Bump composition: d. Bump size:	
24. Header Material (if applicable):	
25. Thermal Resistance: a. θ_{JA} °C/W (approx): b. θ_{JC} °C/W (approx): c. Special thermal dissipation construction techniques:	
26. Test circuits, bias levels, & operational conditions imposed during the supplier's life and environmental tests:	See attached ☐ Not available ☐
27. Fault Grade Coverage (%)	% Not digital circuitry ☐
28. Maximum Process Exposure Conditions: a. MSL @ rated SnPb temperature: b. MSL @ rated Pb-free temperature: c. Maximum dwell time @ maximum process temperature:	** Note: Temperatures are as measured on the center of the plastic package body top surface.* _____ at _____ °C (SnPb) _____ at _____ °C (Pb-free) _____

Attachments:		Requirements:
Die Photo	☐	1. A separate Certification of Design, Construction & Qualification must be submitted for each P/N, wafer fab, and assembly location.
Package Outline Drawing	☐	
Die Cross-Section Photo/Drawing	☐	
Wire Bonding Diagram	☐	2. Design, Construction & Qualification shall be signed by the responsible individual at the supplier who can verify the above information is accurate and complete. Type name and sign below.
Die Placement Diagram	☐	
Test Circuits, Bias Levels, & Conditions	☐	

Completed by: Date:	Certified by: Date:
Typed or Printed: Signature: Title:	

Appendix 3: Plastic Package Opening for Wire Bond Testing

A3.1 Purpose

The purpose of this Appendix is to define a guideline for opening plastic packaged devices so that reliable wire pull or bond shear results will be obtained. This method is intended for use in opening plastic packaged devices to perform wire pull testing after temperature cycle testing or for bond shear testing.

A3.2 Materials and Equipment

A3.2.1 Etchants

Various chemical strippers and acids may be used to open the package dependent on your experience with these materials in removing plastic molding compounds. Red Fuming Nitric Acid has demonstrated that it can perform this function very well on novolac type epoxies, but other materials may be utilized if they have shown a low probability for damaging the bond pad material.

A3.2.2 Plasma Strippers

Various suitable plasma stripping equipment can be utilized to remove the plastic package material.

A3.3 Procedure

a. Using a suitable end mill type tool or dental drill, create a small impression just a little larger than the chip in the top of the plastic package. The depth of the impression should be as deep as practical without damaging the loop in the bond wires.

b. Using a suitable chemical etchant or plasma etcher, remove the plastic material from the surface of the die, exposing the die bond pad, the loop in the bond wire, and at least 75% of the bond wire length. Do not expose the wire bond at the lead frame (these bonds are frequently made to a silver plated area and many chemical etchants will quickly degrade this bond making wire pull testing impossible).

c. Using suitable magnification, inspect the bond pad areas on the chip to determine if the package removal process has significantly attacked the bond pad metallization. If a bond pad shows areas of missing metallization, the pad has been degraded and shall not be used for bond shear or wire pull testing. Bond pads that do not show attack can be used for wire bond testing.

Appendix 4: Minimum Requirements for Qualification Plans and Results

The following information is required as a minimum to identify a device that has met the requirements of AEC-Q100 (see Appendix Templates 4A and 4B). Submission of data in this format is optional. However, if these templates are not used, the supplier must ensure that each item on the template is adequately addressed. The templates can be downloaded from the AEC website at http://www.aecouncil.com.

A4.1 Plans

1. Part Identification: Customer P/N and supplier P/N.
2. Site or sites at which life testing will be conducted.
3. List of tests to be performed (e.g., JEDEC method, Q100 method, MIL-STD method, etc.) along with conditions. Include specific temperature(s), humidity, and bias to be used.
4. Sample size and number of lots required.
5. Time intervals for end-points (e.g., 0 hour, 500 hour, 1000 hour, etc.).
6. Targeted start and finish dates for all tests and end-points.
7. Supplier name and contact.
8. Submission date.
9. Material and functional details and test results of devices to be used as generic data for qualification. Include rationale for use of generic data.

A4.2 Results

All of above plus:

1. Date codes and lot codes of parts tested.
2. Process identification.
3. Fab and assembly locations.
4. Mask number or designation.
5. Number of failures and number of devices tested for each test.
6. Failure analyses for all failures and corrective action reports to be submitted with results.

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Appendix Template 4A: AEC-Q100 Qualification Test Plan

Q100G QUALIFICATION TEST PLAN							
USER COMPANY:				DATE:			
USER P/N:				TRACKING NUMBER:			
USER SPEC #:				USER COMPONENT ENGINEER:			
SUPPLIER COMPANY:				SUPPLIER MANUFACTURING SITES:			
SUPPLIER P/N:				PPAP SUBMISSION DATE:			
SUPPLIER FAMILY TYPE:				REASON FOR QUALIFICATION:			
STRESS TEST	ABV	TEST#	TEST METHOD	Test Conditions/S.S. per Lot/# Lots (identify temp, RH, & bias)	REQUIREMENTS S.S	REQUIREMENTS # LOTS	RESULTS Fails/S.S./# lots
Preconditioning	PC	A1	JEDEC J-STD-020	Peak Reflow Temp. = Preconditioning used = ______	Min. MSL = 3		MSL =
Temperature Humidity Bias or HAST	THB / HAST	A2	JESD22-A101/A110		77	3	
Autoclave or Unbiased HAST	AC / UHST	A3	JESD22-A102/A118		77	3	
Temperature Cycle	TC	A4	JESD22-A104		77	3	
Power Temperature Cycling	PTC	A5	JESD22-A105		45	1	
High Temperature Storage Life	HTSL	A6	JESD22-A103		45	1	
High Temperature Operating Life	HTOL	B1	JESD22-A108		77	3	
Early Life Failure Rate	ELFR	B2	AEC Q100-008		800	3	
NVM Endurance, Data Retention, & Operational Life	EDR	B3	AEC Q100-005		77	3	
Wire Bond Shear	WBS	C1	AEC Q100-001		5	1	
Wire Bond Pull Strength	WBP	C2	MIL-STD-883 - 2011		5	1	
Solderability	SD	C3	JESD22-B102		15	1	
Physical Dimensions	PD	C4	JESD22-B100/B108		10	3	
Solder Ball Shear	SBS	C5	AEC Q100-010		10	3	
Lead Integrity	LI	C6	JESD22-B105		5	1	
Electromigration	EM	D1					
Time Dependent Dielectric Breakdown	TDDB	D2					
Hot Carrier Injection	HCI	D3					
Negative Bias Temperature Instability	NBTI	D4	______	______	______		______
Stress Migration	SM	D5	______	______	______		______
Pre- and Post-Stress Electrical Test	TEST	E1	Test to spec				
Electrostatic Discharge Human Body or Machine Model	HBM / MM	E2	AEC Q100-002/3		See Test Method		
Electrostatic Discharge Charged Device Model	CDM	E3	AEC Q100-011		See Test Method		
Latch-Up	LU	E4	AEC Q100-004		6	1	
Electrical Distributions	ED	E5	AEC Q100-009		30	3	
Fault Grading	FG	E6	AEC-Q100-007				
Characterization	CHAR	E7	AEC Q003				
Electrothermally Induced Gate Leakage	GL	E8	AEC Q100-006	For information only	6	1	
Electromagnetic Compatibility	EMC	E9	SAE J1752/3		1	1	
Short Circuit Characterization	SC	E10	AEC Q100-012	______	10	3	______
Soft Error Rate	SER	E11	JESD89-1, -2, -3	______	3	1	______
Process Average Test	PAT	F1	AEC Q001				
Statistical Bin/Yield Analysis	SBA	F2	AEC Q002				
Hermetic Package Tests	MECH	G1-4	Series		39	3	
Package Drop	DROP	G5			5	1	
Lid Torque	LT	G6	MIL-STD-883 - 2024		5	1	
Die Shear Strength	DS	G7	MIL-STD-883 - 2019		5	1	
Internal Water Vapor	IWV	G8	MIL-STD-883 - 1018		3	1	
Supplier:				Approved by: (User Engineer)			

Automotive Electronics Council
Component Technical Committee

Appendix Template 4B: AEC-Q100 Generic Data

Objective:		*Package:*		*Qual Plan Ref #:*	
Device:		*Fab/Assy/Test:*		*Date Prepared:*	
Cust PN:		*Device Engr:*		*Prepared by:*	
Maskset:		*Product Engr:*		*Date Approved:*	
Die Size:		*Component Engr:*		*Approved by:*	

Test #	ABV	Q100 Test Conditions	End-Point Requirements	Sample Size/Lot	# of Lots	Total # Units	Part to be Qualified	Differences from Q100	Generic Family part A	Differences from Q100	Generic Family part B	Differences from Q100
A1	PC	JEDEC J-STD-020	TEST = ROOM	All surface mount parts prior to A2, A3, A4, A5								
A2	THB / HAST	JESD22-A101/A110	TEST = ROOM and HOT	77	3	231						
A3	AC / UHST	JESD22-A102/A118	TEST = ROOM	77	3	231						
A4	TC	JESD22-A104	TEST = HOT	77	3	231						
A5	PTC	JESD22-A105	TEST = ROOM and HOT	45	1							
A6	HTSL	JESD22-A103	TEST = ROOM and HOT	45	1							
B1	HTOL	JESD22-A108	TEST = ROOM, HOT, and COLD	77	3	231						
B2	ELFR	AEC Q100-008	TEST = ROOM and HOT	800	3	2400						
B3	EDR	AEC Q100-005	TEST = ROOM and HOT	77	3	231						
C1	WBS	AEC Q100-001	Cpk>1.5 and in SPC	An appropriate time period for each bonder to be used								
C2	WBP	MIL-STD-883 – 2011										
C3	SD	JESD22-B102	>95% solder coverage	15	1	15						
C4	PD	JESD22-B100/B108	Cpk > 1.5 per JESD95	10	3	30						
C5	SBS	AEC Q100-010	Two 220°C reflow cycles before SBS									
C6	LI	JESD22-B105	No lead breakage or finish cracks	10 leads from each of 5	1	5						
D1	EM											
D2	TDDB											
D3	HCI											
D4	NBTI											
D5	SM											
E1	TEST		All parametric and functional tests	All units	-	All						
E2	HBM / MM	AEC Q100-002/3	TEST = ROOM and HOT	3 per V level per pin combo	1	Var.						
E3	CDM	AEC Q100-011	TEST = ROOM and HOT	3 per V level	1	Var.						
E4	LU	AEC Q100-004	TEST = ROOM and HOT	6	1	6						
E5	ED	AEC Q100-009	TEST = ROOM, HOT, and COLD	30	3	90						
E6	FG	AEC Q100-007	90%									
E7	CHAR	AEC Q003										
E8	GL	AEC Q100-006 (for information only)	TEST = ROOM	6	1	6						
E9	EMC	SAE J1752/3		1	1	1						
E10	SC	AEC Q100-012		10	3	30						
E11	SER	JESD89-1, -2, -3		3	1	3						
F1	PAT	AEC Q001		All units	-	All						
F2	SBA	AEC Q002		All units	-	All						
G1	MS	JESD22-B104	TEST = ROOM	39	3	117						
G2	VFV	JESD22-B103	TEST = ROOM	39	3	117						
G3	CA	MIL-STD-883 – 2001	TEST = ROOM	39	3	117						
G4	GFL	MIL-STD-883 – 1014		39	3	117						
G5	DROP		TEST = ROOM	5	1	5						
G6	LT	MIL-STD-883 – 2024		5	1	5						
G7	DS	MIL-STD-883 – 2019		5	1	5						
G8	IWV	MIL-STD-883 - 1018		3	1	3						

Appendix Template 4B: AEC-Q100 Generic Data (continued)

Part Attributes	Part to be Qualified	Generic Family part A	Generic Family part B
User Part Number			
Supplier Part Number			
Die Fabrication Site			
Package Assembly Site			
Final Test Site			
Die Family (Product Line)			
Fabrication Process Technology			
Die Size (W x L x T)			
Die Metallization			
Die Interlevel Dielectric			
Die Passivation			
Die Preparation/Singulation			
Die Attach Material			
Package Type/Pin Count			
Mold Compound Supplier/ID			
Wire Bond Material/Diameter			
Wire Bond Methods			
Leadframe Material			
Die Pad/Flag Dimensions			
Lead Finish			
Die Header Material			
Operating Supply Voltage Range			
Operating Temperature Range			
Operating Frequency Range			
Analog Features/Blocks			
Digital Features/Blocks			
Type(s) of (Embedded) Memory			
Memory Size(s)			
Other Characteristics			

Appendix 5: Part Design Criteria to Determine Need for EMC Testing

A5.1 Use the following criteria to determine if a part is a candidate for EMC testing:

a. Digital technology, LSI, products with oscillators or any technology that has the potential of producing radiated emissions capable of interfering with communication receiver devices. Examples include microprocessors, high speed digital IC's, FET's incorporating charge pumps, devices with watchdogs, and switch-mode regulator control and driver IC's.

b. All new, requalified, or existing IC's that have undergone revisions from previous versions that have the potential of producing radiated emissions capable of interfering with communication receiver devices.

A5.2 Examples of factors that would be expected to affect radiated emissions:

- Clock drive (internal or external) I/O Drive
- Manufacturing process or material composition that reduces rise/fall times (e.g., lower E dielectric, lower p metallization, etc.)
- Minimum feature size (e.g., die shrink)
- Package or pinout configuration
- Leadframe material

Appendix 6: Part Design Criteria to Determine Need for SER Testing

A6.1 Use the following criteria to determine if a part is a candidate for SER Testing:

a. The part use application will have a significant radiation exposure such as an aviation application or extended service life at higher altitudes.

b. SER testing is needed for devices with large numbers of SRAM or DRAM cells ($\geq$ 1 Mbit). For example: Since the SER rates for a 130 nm technology are typically near 1000 FIT/MBIT, a device with only 1,000 SRAM cells will result in an SER contribution of ~1 FIT.

A6.2 Examples of factors that would be expected to affect SER results:

a. Technology shrink to small $L_{effective}$.

b. Package mold/encapsulate material.

c. Bump material making die to package connections for Flip Chip package applications.

d. Mitigating factors such as implementation of Error Correcting Code (ECC) and Soft Error Detection (SED).

A6.3 Cases where new SER testing may be required:

a. Change in basic SRAM/DRAM transistor cell structure (e.g., L_{eff}, well depth and dopant concentration, isolation method, etc.).

Revision History

Rev #	Date of change	Brief summary listing affected sections
-	June 9, 1994	Initial Release.
A	May 19, 1995	Added copyright statement. Revised sections 2.3, 2.4.1, 2.4.4, 2.4.5, 2.8, 3.2 and 4.2, Tables 2, 3, 4 and Appendix 1, 2. Added Appendix 3.
B	Sept. 6, 1996	Revised sections 1.1, 1.2.3, 2.3, 3.1, and 3.2.1, Tables 2, 3, and 4, and Appendix 2.
C	Oct. 8, 1998	Revised sections 1.1, 1.1.3, 1.2.2, 2.2, 2.3, 2.4.2, 2.4.5, 2.6, 3.1, 3.2.1, 3.2.3, 2.3.4, 4.1, and 4.2, Tables 3 and 4, Appendix 2, and Appendix 3. Added section 1.1.1, Figures 1, 2, and 3, and Test Methods Q100-008 and -009. Deleted sections 2.7 and 2.8.
D	Aug. 25, 2000	Revised sections 1.1 and 2.3, Figures 2, 3, and 4, Tables 2, 3, and 4, Appendix 1, and Appendix 2. Added section 2.3.2, Test Methods Q100-010 and -011, and Figure 1.
E	Jan. 31, 2001	Revised Figure 4.
F	July 18, 2003	Complete Revision.
G	May 14, 2007	Complete Revision. Revised document title to reflect that the stress test qualification requirements are failure mechanism based. Revised sections 1, 1.1, 1.2.1, 1.2.2, 1.2.3, 2.3.1, 2.4.4, 2.5, 3.2, 3.2.3, 4.2, and 4.3, Figure 2, Tables 2 and 3, Appendix 2, Appendix 4A, and Appendix 4B. Added sections 2.1.1, 3.1.1, Table 2 and 3 entries (test #D4, D5, E10, and E11), Appendix 6, and Test Method Q100-012. Deleted Table 2A.